

Instância	Fluxo em Base line	Fluxo em otimizado	Melhora	Instância em Base line	Instância em Otimizado	Melhora (M)	Saiba Base	Saiba VNS	Cenário	Cenário otimizado
ts10Op.psl	1043	958	-8.15%	352	302	-14.20%	6	7	1 5 6 7 11 12 16	1 9 5 13 14 15 7 8
ts20Op.psl	1221	1119	-8.30%	392	383	-2.30%	6	6	1 2 3 7 11 15 16	9 13 14 2 3 7 8
ts30Op.psl	1179	1109	-5.86%	417	413	-0.96%	6	6	1 5 6 10 11 15 16	1 13 14 15 11 12 8
ts40Op.psl	1265	1229	-2.83%	444	443	-0.23%	6	6	1 5 9 10 14 15 16	5 6 7 8 4 16 12
ts50Op.psl	1543	1470	-4.73%	506	481	-4.94%	6	6	1 5 9 10 11 15 16	5 9 10 11 15 16 4
ts60Op.psl	1120	949	-15.27%	352	298	-15.34%	6	7	1 2 3 4 8 12 16	9 13 15 6 7 3 4
ts70Op.psl	1308	1138	-12.93%	418	313	-24.88%	6	6	1 5 6 10 14 15 16	9 1 5 13 14 15 16
ts80Op.psl	1200	1126	-6.17%	449	406	-9.58%	6	6	1 2 6 10 11 15 16	1 9 5 6 7 15 16
ts90Op.psl	1411	1353	-4.11%	498	418	-16.00%	6	6	1 2 6 10 11 15 16	13 14 15 16 4 8 12
ts10Op.psl	1242	1083	-12.80%	482	362	-25.31%	6	6	1 5 6 7 8 12 16	13 14 15 16 4 8 12
ts11Op.psl	2143	1790	-16.47%	609	511	-16.09%	8	8	1 2 7 12 13 18 19 24 25	6 21 16 11 11 2 2 3 4 5
ts12Op.psl	1918	1552	-19.08%	473	420	-11.21%	8	8	1 2 3 4 9 15 20 25	6 7 12 13 14 19 24 25 6 7
ts13Op.psl	2369	2132	-9.97%	667	603	-9.74%	8	8	1 6 7 8 13 14 15 20 25	21 11 12 13 14 19 20 10
ts14Op.psl	2310	2039	-11.73%	676	494	-26.92%	8	8	1 6 11 12 13 14 18 19 20 25	21 22 17 2 3 8 13 14 15
ts15Op.psl	2271	2036	-10.35%	647	536	-17.16%	8	8	1 6 11 16 17 22 23 24 25	6 1 2 2 2 17 12 13 14 15
ts16Op.psl	2482	1924	-22.48%	721	508	-29.82%	8	8	1 6 7 8 13 14 19 20 25	11 21 16 6 7 8 3 4 5
ts17Op.psl	2007	1967	-11.21%	507	511	0.79%	8	8	1 6 11 12 17 18 23 24 25	16 6 1 1 2 12 13 24 25
ts18Op.psl	2051	1920	-6.39%	515	503	-2.33%	8	8	1 2 7 12 13 18 19 20 25	11 12 17 18 23 2 3 4 5
ts19Op.psl	2259	2110	-6.60%	661	550	-16.79%	8	8	1 6 11 16 17 22 23 24 25	1 2 2 2 2 2 2 14 15 10 12
ts20Op.psl	2434	2187	-9.74%	659	540	-18.50%	8	8	1 2 3 8 9 18 19 24 25	6 13 11 21 22 23 18 19 20
ts21Op.psl	2445	2120	-12.84%	669	585	-12.41%	12	12	1 8 15 16 17 24 25 26 31 41 48 49	36 29 43 15 8 2 2 1 3 4 5 6 7
ts22Op.psl	5021	3704	-26.23%	953	782	-17.84%	12	12	1 8 9 10 18 19 25 26 33 34 38 42 49	22 15 36 29 1 4 3 4 4 5 11 12 13 14
ts23Op.psl	5380	4223	-21.65%	1020	833	-18.07%	12	12	1 2 3 10 11 18 19 25 27 38 42 49	43 38 32 15 8 1 1 2 3 1 5 6 3 4
ts24Op.psl	5737	3800	-33.76%	883	632	-28.99%	12	12	1 2 3 10 18 19 25 27 38 42 49	36 8 22 29 43 15 16 17 18 4 5 6 7
ts25Op.psl	4440	3741	-15.74%	858	684	-20.28%	12	12	1 8 15 16 17 24 25 32 33 34 35 42 49	29 8 36 22 1 13 15 16 17 18 20 21
ts26Op.psl	5305	3989	-24.81%	1027	791	-22.57%	12	12	1 2 3 10 11 18 19 25 27 38 42 49	29 38 15 1 13 15 16 17 18 48 20 49
ts27Op.psl	4703	3739	-20.30%	878	719	-18.33%	12	12	1 2 3 4 5 12 19 27 34 41 42 49	1 2 3 4 15 25 45 47 11 12 34 41 42
ts28Op.psl	4528	3584	-20.85%	910	684	-24.74%	14	14	1 8 9 16 23 24 25 26 27 38 42 49	15 26 29 43 1 2 3 4 4 5 46 47 20 27 28
ts29Op.psl	4344	3897	-12.59%	897	812	-9.56%	12	12	1 2	81 8 15 10 31 18 31 24 25 26 27
ts30Op.psl	4817	3827	-20.77%	927	787	-15.55%	12	12	1 2 8 15 16 17 24 25 34 45 46 47 48 49	22 8 34 37 32 33 34 35 36 37 38 39 40
ts31Op.psl	9254	8915	-3.66%	1349	1306	-3.19%	18	18	11 12 22 32 33 39 43 53	81 82 83 84 6 4 7 4 15 35 36 37 38 59 60 30
ts32Op.psl	9151	7922	-13.43%	1225	1144	-6.51%	18	18	1 2 3 4 5 15 25 35 45 55 65 75 76 77 78 79 80 90 100	81 31 21 41 61 11 11 19 92 52 53 54 55 56 57 78 79 78 90 100
ts33Op.psl	17840	11809	-33.68%	1162	1189	2.36%	18	18	1 2 3 4 5 15 18 19 25 34 45 46 47 48 49	41 42 43 43 23 33 34 34 35 36 37 38 39 40 100
ts34Op.psl	8979	7603	-15.32%	1193	1076	-9.81%	18	18	1 2 12 13 14 24 26 35 36 37 38 39 49 59 69 79 89 99 100	61 2 22 32 82 83 83 83 43 13 14 15 16 17 18 19 99 100
ts35Op.psl	9108	8639	-5.15%	1200	1341	-3.95%	18	20	11 11 21 31 41 51 61 62 63 73 74 75 76 87 88 90 100	1 2 3 4 14 94 95 75 35 65 86 36 56 27 67 68 68 48 50
ts36Op.psl	8685	7951	-7.19%	1030	913	-16.24%	18	18	11 11 21 31 32 33 43 53 64 74 75 85 95 96 97 98 99 100	1 31 32 33 34 54 24 64 65 66 16 17 47 48 50 100 100
ts37Op.psl	9044	8537	-5.63%	1204	1064	-10.80%	18	19	1 2 3 4 5 6 7 11 18 19 20 29 39 49 59 69 79 80 90 100	71 11 51 51 52 52 53 54 54 54 55 55 56 56 57 28 29 39 100
ts38Op.psl	6878	7787	-12.29%	1135	998	-11.99%	18	18	11 21 31 32 33 43 53 54 55 65 75 86 86 97 98 99 100	41 71 61 61 21 51 51 52 12 13 33 33 34 35 55 56 97 98 100
ts39Op.psl	8698	8064	-7.29%	1217	1127	-7.81%	18	18	11 21 31 41 42 43 53 63 64 65 66 76 87 88 89 90 100	71 41 61 31 11 51 51 21 22 23 23 53 53 54 55 96 97 97 58 59 60 70
ts40Op.psl	12787	11688	-8.62%	1297	1168	-10.16%	18	18	11 21 22 23 24 34 35 45 46 56 66 67 77 78 88 89 90 100	41 51 17